Xinyang Han

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I design benchmarks for long-horizon, economically valuable work by AI agents, and study how agents can close the capability gaps those benchmarks reveal.

Education

University of California, Berkeley — **Berkeley, CA**
PhD Student | Berkeley Artificial Intelligence Research (BAIR) — *Sep. 2024 to Present*

University of Chinese Academy of Sciences — **Beijing, China**
Computer Science Major — *2020 to 2024*
Concurrent Student in Computer Science, UC Berkeley — *Jan. to Sep. 2023*

Selected Research

Agents' Last Exam (ALE) — ***Co-Lead***

Co-leading a living benchmark that tests whether AI agents can reliably perform economically valuable work across real professional domains, not just short or synthetic tasks.

ALE converts expert-completed projects into long-horizon GUI+CLI workflows with verifiable outcomes; built with **300+ experts**, it spans **1,500+ tasks** across **55 sub-industries** and was featured as the first figure in the opening section of OpenAI's GPT-5.6 release.

JobBench: Aligning Agent Work With Human Will — ***Co-author***

Most occupational-agent benchmarks prioritize economic value; JobBench instead asks which duties workers actually want delegated, orienting evaluation toward human augmentation rather than replacement.

Grounded in ratings from **1,500+ workers**, it spans **130 file-rich tasks** across **35 occupations** with fact-anchored rubric chains and appears in official evaluation reports for Meta's Muse Spark 1.1 and Moonshot's Kimi K3.

MLS-Bench: Can AI Systems Build Better AI? — ***Core Contributor***

Core contributor to a benchmark testing whether agents can invent ML methods that generalize across controlled settings and scales.

Across **140 tasks** in **12 domains**, experiments show that tuning is easier than method invention and that more search, compute, or context alone does not remove the scientific-insight bottleneck.

SAP3D: Sparse-View 3D Reconstruction — ***First Author***

Sparse-view 3D reconstruction must infer both geometry and camera pose from only a few unposed images, where single-view priors and classical multi-view optimization each fall short.

Developed SAP3D to refine camera poses and adapt a view-conditioned diffusion model at test time, improving reconstruction as more views become available; selected as a **CVPR 2024 Highlight**.

U2Seg: Unsupervised Universal Image Segmentation — ***Co-first Author***

Unsupervised segmentation previously required separate systems for semantic, instance, and panoptic tasks, with no unified model spanning all three.

U2Seg learns from self-supervised pseudo-labels in one framework, improving specialized baselines by **+2.6 box AP** on COCO and **+7.0 PixelAcc** on COCOStuff while establishing the first unsupervised panoptic baseline; **CVPR 2024**.

Multimodal AI for Cancer Risk Prediction — ***PhD Research***

Developing multimodal modeling and evaluation methods for cancer risk prediction with physician collaborators.

The research emphasizes clinically meaningful endpoints, calibration, and robustness across clinical settings.

Selected Publications

Agents' Last Exam
arXiv, 2026. Yiyou Sun, **Xinyang Han**†, Weichen Zhang, Yuanbo Pang, Tianyu Wang, Yuhan Cao, Yixiao Huang, et al.

JobBench: Aligning Agent Work With Human Will
arXiv, 2026. Yuetai Li, Yichen Feng, Zhangchen Xu, Zixian Ma, …, **Xinyang Han**, et al.

MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI
ICML 2026 AI for Math Workshop. Bohan Lyu, Yucheng Yang, Siqiao Huang, Jiaru Zhang, Qixin Xu, Xinghan Li, **Xinyang Han**‡, et al.

The More You See in 2D, the More You Perceive in 3D
CVPR 2024 Highlight. **Xinyang Han***, Zelin Gao*, Angjoo Kanazawa, Shubham Goel, Yossi Gandelsman.

Unsupervised Universal Image Segmentation
CVPR 2024. Dantong Niu*, Xudong Wang*, **Xinyang Han***, Long Lian, Roei Herzig, Trevor Darrell.

MuNeRF: Robust Makeup Transfer in Neural Radiance Fields
IEEE TVCG, 2025. Yujie Yuan*, **Xinyang Han***, Yue He, Fanglue Zhang, Lin Gao.

* *Equal contribution.* † *Co lead.* ‡ *Core contributor.*

Leadership & Academic Service

AI Investment Summit 2025 at UC Berkeley *Founding Organizer*
Orchestrated a **100+ member** cross-campus team to deliver a summit for approximately **1,000 attendees**, featuring UC Berkeley Chancellor Rich Lyons, Sequoia partner Konstantine Buhler, and Tianfu Fu, a GPT-5 core contributor.

Workshop on Agent Behavior at COLM 2026 *Organizer*
Convening research on how AI agents behave, fail, and interact in deployed settings; supported by Anthropic, Google, and OpenAI.

Conference Organization & Peer Review *Academic Service*
Communications Chair, CHIL 2026; Logistics Chair, CHIL 2025. Reviewer for ICLR, NeurIPS, KDD, and CHIL 2025.

Selected Talks & Seminars

Agents' Last Exam: Evaluating AI Agents on Real Work *June to July 2026*
Snorkel AI; UC Berkeley; Kimi AI; WebAgentLab; Valkyrie AI

3D Foundation Models for Generation and Reconstruction *July 2024*
ICT Turing Class

Technical Skills

Agent Evaluation	computer-use environments; long-horizon tasks; rubric design; artifact grading; evaluation harnesses; model benchmarking
Programming & ML	Python; C++; PyTorch; JAX; TensorFlow
Computer Vision	representation learning; sparse-view 3D reconstruction; unsupervised image segmentation